

A Six-Parameter Extension of the Direct Somos–4 Algebraic Generating Function

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Abstract

The preceding paper constructs a five-parameter quadratic algebraic generating function with prescribed initial Hankel determinants (λ, m, r, η) and Somos–4 parameter τ . We adjoin a second Somos coefficient α . If $q^6 = \alpha$, the weighted transport

$$G_{\alpha; \lambda, m, r, \eta, \tau}(x) = G_{1; \lambda, q^{-2}m, q^{-6}r, q^{-12}\eta, q^{-8}\tau}(qx)$$

preserves the four prescribed ordinary Hankel determinants and changes the recurrence to

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2.$$

The once-shifted determinants acquire an explicit grading factor. We also transport the discriminant exactly and show that its residual genus-one quartic has the rational model

$$Y^2 = (1 - (A + \alpha B)z + \tau z^2)^2 - 4\alpha^2 z^3,$$

where A and B are rational functions of the initial Hankel data. This separates curve-level descent from descent of the distinguished formal-power-series branch. The latter remains open; no nonexistence claim is made.

1 Introduction

The five-parameter construction in [1] uses the notation

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta)$$

and the recurrence parameter τ in

$$H_{k+4}H_k = H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{1}$$

We retain that notation throughout. The new parameter α is the coefficient of the first bilinear term, so the six-parameter recurrence is

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{2}$$

Thus the parameter list is $(\lambda, m, r, \eta, \alpha, \tau)$, and specialization $\alpha = 1$ recovers the normalization of [1].

The extension is obtained by a weighted scaling over a field containing a sixth root of α . The determinant identities and the transported discriminant are exact formal identities. They do not require expansion of the lengthy quadratic polynomial from the preceding paper. What is not settled by these identities is whether the distinguished normalized series itself descends from the sixth-root extension to $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

2 The five-parameter input

Let

$$G_{1;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} g_k x^k$$

be the selected algebraic series of [1]. Define

$$H_N(g) = \det(g_{i+j})_{0 \leq i,j \leq N-1}, \quad D_N(g) = \det(g_{i+j+1})_{0 \leq i,j \leq N-1}.$$

The normalization in the preceding paper is

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta), \quad D_N = H_{N+2}. \quad (3)$$

Its quadratic discriminant factors as

$$\Delta_1(x) = (x\lambda^2\eta mr^2)^2 \left((1 - Jx + \tau x^2)^2 - 4x^3 \right), \quad (4)$$

where

$$J = \frac{\lambda\eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \quad (5)$$

These statements are the only inputs from the five-parameter construction needed below.

3 Weighted transport

Work over

$$\mathbb{K} = \mathbb{Q}(\lambda, m, r, \eta, q, \tau), \quad q^6 = \alpha,$$

and introduce the primed parameters

$$\lambda' = \lambda, \quad m' = q^{-2}m, \quad r' = q^{-6}r, \quad \eta' = q^{-12}\eta, \quad \tau' = q^{-8}\tau. \quad (6)$$

Define

$$G_{\alpha;\lambda,m,r,\eta,\tau}(x) := G_{1;\lambda',m',r',\eta',\tau'}(qx). \quad (7)$$

If

$$G_{1;\lambda',m',r',\eta',\tau'}(x) = \sum_{k \geq 0} a_k x^k, \quad G_{\alpha;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} b_k x^k,$$

then

$$b_k = q^k a_k. \quad (8)$$

Proposition 1 (Determinant weights). *For every $N \geq 1$,*

$$H_N(b) = q^{N(N-1)} H_N(a), \quad D_N(b) = q^{N^2} D_N(a). \quad (9)$$

Proof. In the ordinary Hankel matrix, its (i, j) entry is multiplied by q^{i+j} . Factoring q^i from row i and q^j from column j gives the exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$. For the shifted matrix the entry is multiplied by q^{i+j+1} . The additional factor q in every row contributes N , so the total exponent is $N(N-1) + N = N^2$. $\square$

4 The two-coefficient Somos–4 recurrence

Theorem 2. *Let $\widehat{H}_N = H_N(b)$. Then*

$$(\widehat{H}_1, \widehat{H}_2, \widehat{H}_3, \widehat{H}_4) = (\lambda, m, r, \eta)$$

and

$$\widehat{H}_{k+4}\widehat{H}_k = \alpha\widehat{H}_{k+3}\widehat{H}_{k+1} + \tau\widehat{H}_{k+2}^2, \quad k \geq 1. \quad (10)$$

Proof. Equation (9) and (6) give

$$\widehat{H}_1 = \lambda, \quad \widehat{H}_2 = q^2 m' = m, \quad \widehat{H}_3 = q^6 r' = r, \quad \widehat{H}_4 = q^{12} \eta' = \eta.$$

Write $H'_N = H_N(a)$. The primed sequence satisfies

$$H'_{k+4}H'_k = H'_{k+3}H'_{k+1} + \tau'H_{k+2}^2.$$

The weight difference between the left-hand product and the first product on the right is 6, while that between the left-hand product and H_{k+2}^2 is 8. Hence

$$\widehat{H}_{k+4}\widehat{H}_k = q^6\widehat{H}_{k+3}\widehat{H}_{k+1} + q^8\tau'\widehat{H}_{k+2}^2.$$

Now use $q^6 = \alpha$ and $q^8\tau' = \tau$. □

Corollary 3 (Low-order recurrence check). *The first two values forced beyond the initial tuple are*

$$\widehat{H}_5 = \frac{\alpha\eta m + \tau r^2}{\lambda}, \quad \widehat{H}_6 = \frac{\alpha\widehat{H}_5 r + \tau\eta^2}{m}.$$

These formulas provide a direct indexing check for any coefficient expansion of (7).

5 The transported companion normalization

Proposition 4. *If the primed series obeys $D_N(a) = H'_{N+2}$, then*

$$\widehat{D}_N := D_N(b) = q^{-3N-2}\widehat{H}_{N+2}, \quad N \geq 1. \quad (11)$$

Proof. By (9),

$$D_N(b) = q^{N^2}H'_{N+2}, \quad \widehat{H}_{N+2} = q^{(N+2)(N+1)}H'_{N+2}.$$

Their exponent difference is $N^2 - (N+2)(N+1) = -3N-2$. □

Remark 5. A constant coefficient rescaling $b_k \mapsto cb_k$ changes the ratio $D_N(b)/\widehat{H}_{N+2}$ by c^{-2} . It cannot remove the factor q^{-3N} . Thus the literal identity $D_N = H_{N+2}$ is not invariant under this graded transport. Equation (11), rather than the ungraded identity, is the companion normalization naturally inherited from the five-parameter series.

6 Low-order expansion and non-descent of the transported branch

For the five-parameter selected series, the companion normalization gives $g_0 = \lambda$ and $g_1 = r$. Applied to the primed parameters in (6), this yields $a_0 = \lambda$ and $a_1 = r' = q^{-6}r$. Therefore (8) gives

$$b_0 = \lambda, \quad b_1 = q^{-5}r = \frac{qr}{\alpha}. \quad (12)$$

The next two coefficients obtained by direct expansion are

$$b_2 = \frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} = \frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda}, \quad (13)$$

$$b_3 = \frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2}. \quad (14)$$

Hence

$$\begin{aligned} G_\alpha(x) &= \lambda + \frac{qr}{\alpha}x + \left(\frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda} \right) x^2 \\ &\quad + \left(\frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2} \right) x^3 + O(x^4). \end{aligned} \quad (15)$$

The first coefficient also follows from (11) at $N = 1$:

$$b_1 = D_1(b) = q^{-5}H_3(b) = q^{-5}r.$$

Thus the fractional power is forced by the transported companion data rather than caused by an insufficient simplification.

Put

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau), \quad L = K(q), \quad q^6 = \alpha.$$

Proposition 6 (Non-descent of the transported branch). *Regard α as an indeterminate. In the generic nondegenerate case $r \neq 0$, the transported series (7) does not belong to $K[[x]]$.*

Proof. Its coefficient of x is $b_1 = q^{-5}r = qr/\alpha$, which is not in K for generic α . Equivalently, if $\sigma(q) = \zeta q$ with $\zeta^6 = 1$ and $\zeta \neq 1$, then $\sigma(b_1) = \zeta^{-5}b_1 = \zeta b_1 \neq b_1$. $\square$

Although the coefficients do not descend, the fractional powers cancel in the ordinary Hankel determinants. For example,

$$H_2(b) = b_0b_2 - b_1^2 = \lambda \left(\frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} \right) - q^{-10}r^2 = m.$$

Together with the theorem above, this establishes the precise distinction: ordinary Hankel descent holds, whereas descent of the selected transported series fails.

7 Exact transport of the discriminant

Separate the old invariant into

$$A = \frac{\lambda\eta}{mr} + \frac{\tau mr}{\lambda\eta}, \quad B = \frac{m^2}{\lambda r} + \frac{r^2}{m\eta}. \quad (16)$$

After substitution of (6),

$$J' = q^{-4}A + q^2B. \quad (17)$$

Consequently the substitution $x \mapsto qx$ gives

$$qJ' = q^{-3}A + q^3B = \alpha^{-1/2}A + \alpha^{1/2}B. \quad (18)$$

Proposition 7 (Transported discriminant). *The discriminant obtained by substituting (6) and $x \mapsto qx$ into (4) is*

$$\Delta_\alpha(x) = q^{-50}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (q^{-3}A + q^3B)x + q^{-6}\tau x^2 \right)^2 - 4q^3x^3 \right]. \quad (19)$$

Equivalently,

$$\Delta_\alpha(x) = \alpha^{-25/3}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (\alpha^{-1/2}A + \alpha^{1/2}B)x + \frac{\tau}{\alpha}x^2 \right)^2 - 4\alpha^{1/2}x^3 \right]. \quad (20)$$

Proof. The square prefactor has total q -weight $2(1 - 12 - 2 - 12) = -50$. The three terms in the residual factor transform as $J'x \mapsto qJ'x$, $\tau'x^2 \mapsto q^2\tau'x^2 = q^{-6}\tau x^2$, and $x^3 \mapsto q^3x^3$. Equations (17) and $q^6 = \alpha$ give the stated forms. $\square$

The overall monomial factor in (19) is a square in $\mathbb{K}(x)$ and does not affect the residual double cover. It should, however, be retained when comparing discriminants of particular polynomial representatives, because multiplying a quadratic equation by a scalar squares that scalar in its discriminant.

8 A rational curve-level model

Remove the square prefactor in (20) and put

$$x = \alpha^{1/2}z. \quad (21)$$

The residual equation becomes

$$Y^2 = \left(1 - (A + \alpha B)z + \tau z^2\right)^2 - 4\alpha^2 z^3. \quad (22)$$

All coefficients of (22) belong to $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

For comparison with the quartic convention used in [1], set $X = z^{-1}$ and $Y = z^{-2}y$. Then (22) is birational to

$$y^2 = \left(X^2 - (A + \alpha B)X + \tau\right)^2 - 4\alpha^2 X. \quad (23)$$

Thus the curve descends to the rational six-parameter field. The coordinate change (21) nevertheless rescales coefficients of a formal power series and does not preserve the normalized Hankel data. Curve descent and descent of the distinguished generating series are therefore different problems.

9 Hankel representatives and Galois-equivariant transforms

Preservation of the ordinary Hankel transform alone does not characterize the distinguished algebraic series. For a nondegenerate Hankel sequence define

$$\beta_n = \frac{H_{n-1}H_{n+1}}{H_n^2}, \quad H_0 = 1. \quad (24)$$

All H_n generated by (10), and hence all β_n , lie in K . For arbitrary $a_0, a_1, \dots \in K$, the Jacobi fraction

$$\tilde{G}(x) = \frac{H_1}{1 - a_0x - \frac{\beta_1 x^2}{1 - a_1x - \frac{\beta_2 x^2}{1 - a_2x - \ddots}}} \quad (25)$$

has ordinary Hankel determinants H_n . Thus radical-free formal-series representatives over K exist, but they need not be quadratic algebraic, preserve the graded companion identity, or arise from the distinguished Stieltjes splitting.

The following observation excludes a broad class of attempted rationalizations.

Proposition 8 (Galois-equivariant obstruction). *Let $T : L[[x]] \rightarrow L[[x]]$ be injective and commute with every K -automorphism of L . If $T(G) \in K[[x]]$, then $G \in K[[x]]$. If T is invertible, the two conditions are equivalent.*

Proof. For every $\sigma \in \text{Gal}(L/K)$, $T(\sigma G) = \sigma T(G) = T(G)$. Injectivity gives $\sigma G = G$, so G is fixed by the Galois action and belongs to $K[[x]]$. $\square$

For $c \in K$, the binomial transform

$$B_c(G)(x) = \frac{1}{1-cx} G\left(\frac{x}{1-cx}\right)$$

is defined over K , commutes with the Galois action, and is invertible with inverse B_{-c} . Likewise, the invert transform

$$I_c(G)(x) = \frac{G(x)}{1-cxG(x)}$$

is defined over K , Galois-equivariant, and invertible with inverse I_{-c} . Consequently neither transform can turn the non-descending transported branch into a series in $K[[x]]$. This conclusion concerns these transforms over the base field; it does not exclude a new algebraic construction with different companion data.

10 The remaining open problem

The descent question for the specific transported branch is now settled negatively in the generic case by (12). The broader question is different:

Does there exist a direct quadratic algebraic generating function over $K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$ whose ordinary Hankel determinants have initial values (λ, m, r, η) and satisfy (10)?

Jacobi reconstruction gives many formal representatives over K , so no nonexistence statement is possible for the unrestricted class of all formal series with the same ordinary Hankel transform. A substantive existence or nonexistence theorem must additionally specify algebraicity, a companion normalization, a finite universal transform, or a distinguished continued-fraction splitting.

In particular, it remains open whether there is a quadratic algebraic series over K with the same ordinary Hankel orbit and a suitably modified, radical-free companion normalization. Possible approaches include a direct six-parameter tail recurrence over K and rational splittings of modified companion products. These are prospective constructions, not assumptions in the results proved here.

11 Conclusion

The weighted substitution (7) gives an exact six-parameter extension of the direct Somos-4 construction. The prescribed ordinary Hankel tuple is unchanged, the recurrence becomes (10), and the companion determinants satisfy the graded identity (11). The explicit expansion (15) shows that the selected transported branch genuinely requires the sixth-root extension in the generic case, already because $b_1 = q^{-5}r$. Ordinary Hankel descent therefore does not imply selected-series descent.

Exact transport of the discriminant gives (19); after a coordinate rescaling, its residual genus-one curve has the rational model (22). Hence curve-level descent and selected-series descent are distinct. Standard binomial and invert transforms defined over the rational parameter field cannot create descent, by Galois equivariance and injectivity. The remaining mathematically substantive problem is to construct a different direct quadratic algebraic representative over K , or to prove nonexistence within a precisely specified Hankel-compatible algebraic class.

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The manuscript was revised with assistance from AI-based language and symbolic calculation tools. Their use was directed toward notation control, LaTeX editing, and exact algebraic checks. The author remains responsible for the mathematical statements and the final text.

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